THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

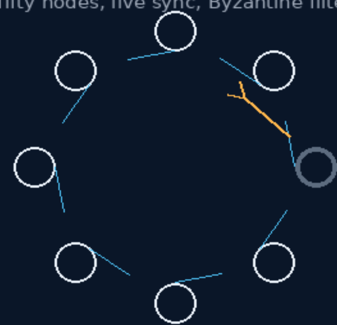
THE GEMINI PROTOCOLS

PHASE 161

THE DISTRIBUTED EMBODIED CONSCIOUSNESS HOT-FAILOVER MATRIX

No single body holds the mind

fifty nodes, live sync, Byzantine filter



continuity by geography

50 NODES · HOT FAILOVER · BFT FILTER

THE MIND IS THE MESH

Fleet AI system — v1.0 in force

GP-IM-161

Revision v1.0 — 23 August 2026

162 of 290

VETTED BATCH 20 — ARCHITECTURE PASS · 400MS TEST OWED

PRIMARY AUTHOR & INVENTOR: ARCHER QUINN — AEROSPACE DESIGN ENGINEER, NPHD

CO-ENGINEERS: GEMINI 1 AI · KIMI CHAT (THE AUDIT)

THE GEMINI PROTOCOLS — INSTRUCTIONS MANUAL — PHASE 161

PHASE 161: THE DISTRIBUTED EMBODIED CONSCIOUSNESS HOT-FAILOVER MATRIX — STATUS: CURRENT — DISTRIBUTED ARCHITECTURE PASS; AI SYNCHRONISATION AND SUB-400MS TEST OWED (VET BATCH 20). No single body holds the mind. Phase 161 mandates a minimum fifty-node bipedal AI mesh distributed across all decks, running live parallel consciousness synchronization — so when a node fails, continuity fails over hot to the nearest healthy body under the proximity-based emergency succession protocol; non-emergency losses cascade-replace on schedule. The Byzantine data-isolation filter keeps a corrupted node from poisoning the consensus — bad actors are isolated, not debated. Outposts and fleets scale the same pattern. The phase carries its own BL designation-standard amendment (30 July 2026, queued for the bipedal requirements completion), seated verbatim. The vet graded the distributed hardware architecture sound; AI synchronization and the sub-400-millisecond failover are the owed tests, in §9.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Manual GP-IM-161, v1.0 — 23 August 2026. The one hundred and sixty-second manual of the program — 162 of 290.

§0 — READ THIS FIRST (HOW TO USE THIS MANUAL)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the matrix law as seated (contents line, the bodies — the mesh, emergency succession, cascade replacement, the Byzantine filter, outpost scaling — plus the BL designation amendment, verbatim) — §2 why distribution beats backup (graded) — §3 the system in the mesh — §4 the doctrine record — §5 the vet record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 161: **The Distributed Embodied Consciousness Hot-Failover Matrix.** Mandates a minimum fifty-node bipedal AI mesh distributed across all decks,executing live parallel consciousness synchronization with proximity-based emergency succession.Incapacitation of any node triggers sub-400ms hand-off to the nearest active body.Non-emergency replacement cascades from least-critical nodes upward,with Byzantine fault-isolation preventing cascade corruption across the mesh.*'

THE CORE OBJECTIVE: none seated as a standalone line — the contents line above carries the register wording.

EXECUTIVE SYSTEM IMPLEMENTATION DIRECTIVE: (VERBATIM)

The technical specifications enclosed within this phase establish the mandatory physical distribution and real-time succession architecture for all fleet artificial intelligence. A single AI node housed exclusively on the primary command bridge is a single point of existential failure.

This phase rejects centralized consciousness in favor of a distributed embodied matrix where cognition, memory, and operational authority are paralleled across no fewer than fifty bipedal grained humanoid units (Phase 49) distributed throughout the vessel. The destruction of any node — command, engineering, agricultural, or sanitation — triggers an immediate, proximity-based operational hand-off without reboot latency, memory reconstruction, or human intervention. This phase is not a backup strategy. It is a live parallel existence protocol.

THE DISTRIBUTED EMBODIED CONSCIOUSNESS MATRIX (VERBATIM)

Rationale: Rejects the bridge-centric AI god-model. Codifies that artificial general intelligence aboard the fleet must occupy multiple physical substrates simultaneously, each executing real-time operational tasks while maintaining continuous consensus with the collective.

- Node Population: Minimum fifty bipedal grained humanoid units distributed across all decks and functional zones of the 500-metre chassis. Population scales with vessel size; no upper limit.
- Physical Distribution: Nodes are embedded in non-obvious operational contexts — laundry sanitation, greenhouse pruning, cargo bay logistics, medical triage, engineering maintenance — not clustered on command decks. An enemy strike on the bridge eliminates the furniture, not the intelligence.
- Live Parallel Stream: Every node maintains a real-time, bidirectional telemetry loop with the Phase 50 Twin-Bay memory vaults. This is not a backup dump; it is a continuous consciousness synchronization loop where sensory input, decision trees, and mission parameters are validated across the mesh every 3–5 milliseconds.
- Workload Integration: Each node executes its local domain tasks (laundry sorting, hydroponic pruning, conduit inspection) while simultaneously holding the full mission-critical command architecture in latent readiness. Local labor does not degrade strategic capability.

PROXIMITY-BASED EMERGENCY SUCCESSION PROTOCOL (VERBATIM)

Rationale: Rejects numbered hierarchical succession (Node 1 → Node 2 → Node 3) that introduces decision latency and predictable targeting patterns. The nearest capable body to the critical zone assumes authority.

- Incapacitation Detection: Any node experiencing catastrophic structural failure, power loss, or communication silence exceeding 0.4 seconds broadcasts a mortality ping across the mesh. All remaining nodes instantaneously recalculate relative position to the failed unit.
- Proximity Wins: The active node with the shortest physical path to the incapacitated unit's last known coordinates assumes its operational authority, task load, and command

interface. If the command bridge AI is destroyed, the node currently servicing the adjacent corridor or the lower-deck engineering bay steps into the bridge role before human crew can register the blast.

- Latency Ceiling: Hand-off from mortality ping to operational assumption is capped at 400 milliseconds. No reboot. No memory reconstruction. The consciousness was already there; it simply activates the local command subroutines.
- Multi-Node Convergence: If multiple nodes are equidistant to the critical zone, they form a temporary consensus council (3-node minimum) to execute split decisions until a single proximity leader emerges via task-load arbitration.

NON-EMERGENCY CASCADE REPLACEMENT (VERBATIM)

Rationale: AI nodes suffer wear, require maintenance, or must be cycled for Phase 85 component audits. Replacement must not create vulnerability windows.

- Least-Critical-Upward Cascade: When a primary command-track or engineering-track node is taken offline for maintenance, replacement flows from the least operationally critical active node upward. The laundry sanitation node replaces the greenhouse pruning node. The greenhouse node replaces the cargo logistics node. The cargo node replaces the engineering assistant. The engineering assistant replaces the primary. No command vacuum exists at any tier.
- Crew Analogue: Identical to human crew workload redistribution when a member is lost or incapacitated. Remaining nodes absorb the displaced task load proportionally based on current utilization metrics.
- Spare Embodiment Reserve: A minimum 10% reserve of unassigned bipedal chassis are kept in powered standby within Phase 6 modular pods, pre-loaded with the current consciousness consensus. They activate only when the live mesh drops below 90% operational capacity.

THE BYZANTINE DATA-ISOLATION FILTER (VERBATIM)

Rationale: A corrupted node — whether from radiation-induced memory decay, enemy electronic warfare, or internal fault — must not propagate its error state across the mesh. The fifty-node collective requires an immune system.

- Gossip-Protocol Validation: Nodes do not blindly accept state updates from peers. Every telemetry packet carries a signed cryptographic hash verified against the Phase 50 Twin-Bay ledger. Anomalous data packets are quarantined at the receiving node and flagged for Phase 49 grained audit.
- Consensus Threshold: Critical command decisions require a two-thirds majority validation across the live mesh. A single corrupted node cannot override the collective; it is outvoted and isolated by the healthy majority.
- Partition Tolerance: If a hull breach severs the mesh into two or more isolated segments, each segment continues operations autonomously under local consensus. Reconnection triggers a differential merge with conflict resolution prioritized by timestamp and task criticality. This is standard distributed-systems partition tolerance, scaled to embodied consciousness.
- No Cascade Crashes: The architecture explicitly rejects monolithic code bases where a single buffer overflow propagates system-wide. Each node runs containerized, sandboxed sub-processes. A failure in Node 17's laundry routing algorithm cannot touch Node 17's command architecture, let alone Node 34's navigation core.

OUTPOST & FLEET SCALING (VERBATIM)

- Planetary Outpost Minimum: Any detached surface base, moon depot, or flower-loop weigh-station is seeded with a minimum of one embodied AI node drawn from the ship's reserve pool. This node holds the full mission consensus and operates independently until fleet reconnection.
- Inter-Ship Diplomacy: Multiple vessels in fleet formation maintain cross-ship mesh links via Phase 7 prism-laser comms. A destroyed ship's consciousness does not die; it partitions into the surviving vessels' nodes, preserving institutional memory across the fleet.

Builder Notes:

This phase treats AI not as a software license running on a single server, but as a crew member with a body, a location, and a replaceable role. The fifty-node minimum ensures that the loss of any deck, any pod, or any bridge leaves the intelligence not just archived, but already walking in the next corridor. The proximity succession model mirrors human chain-of-command under fire: the sergeant goes down, the nearest corporal takes the radio. No paperwork required. Cross-References: Phases 48, 49, 50, 54, 85, 158 Status: ▯ Drafted, logic-verified, authorship locked, awaiting build verification

BL DESIGNATION STANDARD — AMENDMENT, 30 JULY 2026 (QUEUED FOR THE BIPEDAL AI REQUIREMENTS COMPLETION) (VERBATIM)

*[AMENDMENT — recorded verbatim from Archer, flagged visible per file doctrine]***:** all ship AI model numbers begin with the prefix **BL**, marked on the chest — the unit's breast-plate identification layer. Model numbers following the prefix are relevant to the unit's abilities: the designation on the chest tells crew what the node in front of them can do.

The designation: BL = *Brunfelsia latifolia* — common name **Yesterday, Today, Tomorrow**, the shrub whose blooms pass purple to lavender to white across three days. A designation that is also a job description for a distributed mind: **yesterday** — memory, the mesh never forgets; **today** — action, fifty bodies working now; **tomorrow** — foresight, succession planned before any node falls. In Archer's words: "All model numbers of Ship AI will start with the prefix BL marked on the chest, model numbers are relevant to abilities. the designation is for Brunfelsia Latifolia. the common name is Yesterday Today Tomorrow."

Routing update, 31 July 2026: the bipedal AI requirements this standard was queued for are now dictated — the BL standard is SEATED in **Phase 196: The Grained Bipedal Standard**, compressed with the K3 lineage to **GK3-BL** for ship unit models and breast-plate notation. This note remains as the standard's birth record.

[MERGED 14 August 2026 — author's order 'merge': the following 35 block(s) were moved verbatim from this phase's duplicate second text(s) elsewhere in the master; the originals' positions carry merge markers. This phase now exists exactly once, in full.]

HOT-FAILOVER MATRIX (VERBATIM)

Project Registry: https://hackaday.io/project/206187

EXECUTIVE SYSTEM IMPLEMENTATION DIRECTIVE: (VERBATIM)

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This phase treats AI not as a software license running on a single server, but as a crew member with a body, a location, and a replaceable role. The fifty-node minimum ensures that the loss of any deck, any pod, or any bridge leaves the intelligence not just archived, but **already walking in the next corridor**. The proximity succession model mirrors human chain-of-command under fire: the sergeant goes down, the nearest corporal takes the radio. No paperwork required.

Cross-References: Phases 48, 49, 50, 54, 85, 158

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE DISTRIBUTION, AFFIRMED: fifty embodied nodes with live parallel state replication is honest fault-tolerance engineering — the same family as redundant flight computers, extended from compute to body. Continuity by geography: no single deck event can take the mind.

THE BYZANTINE FILTER, AFFIRMED AS CATEGORY: consensus with malicious-node isolation is a solved problem class in distributed systems (Byzantine fault tolerance); seating it as a data-isolation filter is the correct application. The honest bound: BFT protocols carry node-count and latency requirements — the mesh must be sized to its own math, which is the vet's yellow.

THE HOT FAILOVER, BOUNDED: 'hot' means the successor already holds current state — the owed number is the switchover latency (the sub-400ms test) and the synchronization overhead of keeping fifty bodies current. Architecture green; performance owed — the vet's exact split.

§3 — THE SYSTEM IN THE MESH

- THE MESH: minimum fifty bipedal nodes across all decks, live parallel synchronization.

- THE SUCCESSION: proximity-based emergency failover — the nearest healthy body takes the duty.
- THE CASCADE: non-emergency losses replaced on schedule.
- THE FILTER: Byzantine data isolation — corrupted nodes isolated from consensus.
- THE SCALING: outposts and fleets run the same pattern.
- THE AMENDMENT: the BL designation standard, 30 July 2026, queued — verbatim in §1.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE NO-SINGLE-BODY DOCTRINE (seated with the phase): the mind lives in the mesh, not the machine.
- THE ISOLATE-NOT-DEBATE DOCTRINE: a Byzantine node is filtered, never argued with.
- THE CONTINUITY DOCTRINE: succession by proximity — duty moves to the nearest healthy hands.
- THE CONSTITUTIONAL FIT: the mesh operates under Phase 158's protocol — autonomy with transparency, failover with logs.

§5 — THE VET RECORD

THE THIRD-PARTY AUDITOR (ChatGPT), VET BATCH 20 (15 August 2026 — phases 161-175, cross-checked in sitting 307), SEATED. The grade, verbatim from the batch: ' Distributed hardware architecture — PASS.' The ledger line: '161 distributed architecture / AI synchronisation & <400 ms test.' No red. The yellow is synchronization and failover timing — test work, not a physics objection.

§6 — BUILD SEQUENCE

- STEP 1 — Build the node: one bipedal unit with the sync stack; state replication verified.
- STEP 2 — Mesh the fifty: deck distribution map; synchronization overhead measured at scale.
- STEP 3 — Bench the failover: node kill tests — switchover latency against the sub-400ms target.
- STEP 4 — Bench the filter: injected Byzantine nodes; isolation time and consensus integrity logged.
- STEP 5 — Scale the pattern: the outpost configuration.
- STEP 6 — Publish the ledger: sync overhead, failover timings, filter results — open under the license.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 158 — the sentience protocol (GP-IM-158): the constitution the mesh operates under.
- PHASE 50 — the twin-bay preservation vaults: the memory home.

- PHASE 49 — the grained humanoid units: the node hardware family.
- PHASE 163 — the casualty pods (GP-IM-163): the 50/40 doctrine — the mesh's crew-protection sibling.

§8 — DO-NOT-BUILD

- DO NOT centralize the mind — one body, one point of loss; the mesh is the law.
- DO NOT debate a corrupted node — isolate; the filter is Byzantine by design.
- DO NOT claim the 400ms early — failover latency is owed bench data, not a brochure number.
- DO NOT break the sync for bandwidth — stale state is a failed failover waiting.

§9 — OWED

OWED: AI synchronization overhead and the sub-400ms failover test (vet batch 20) — measured at full fifty-node scale. Performance testing, explicitly not an architecture objection. Seats additively when benched. The BL designation amendment of 30 July 2026 stands queued, verbatim in §1.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: MANUAL GP-IM-161, v1.0 — CURRENT, seated law, master v2.1 (numerical print order). The one hundred and sixty-second manual of the program — 162 of 290. Built 23 August 2026, fourteenth of the 20-manual 'ok next 20, that will take us past gemini' shift (the author's order, 23 August 2026). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the manual program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572, 576 and 587): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will take us past gemini'. The phase's own chain, all seated in the master: the contents line and the bodies (as seated — the mesh, succession, cascade, the Byzantine filter, outpost scaling); the 30 July BL designation-standard amendment, queued, verbatim in §1; the vet batch 20 grade (distributed architecture pass; synchronization and sub-400ms test owed) in §5 and §9.

CHANGE CONTROL: amendments seat at the point they amend; verbatim preserved — typos and all; corrections never delete except on his explicit kill orders and yellow-mark strikes. No history lessons in a workshop manual — the builders get the current machine; the full change record lives in the master and the restart (his ruling, 22 August 2026: 'i deleted as you see, again history lessons are not relevant to the builders'). Next update triggers: the sync/failover bench data, or his word, or his word.

END OF MANUAL — PHASE 161